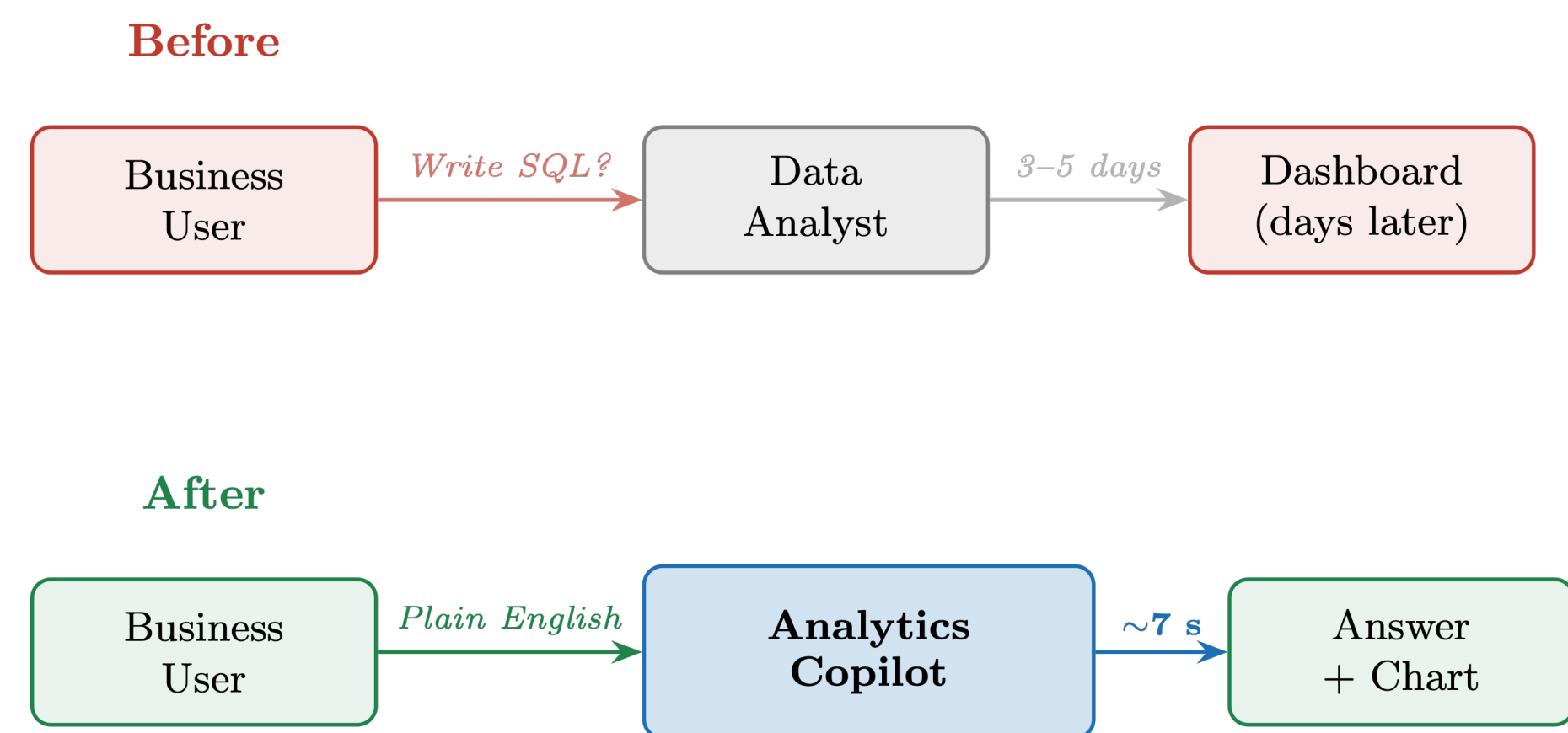


Analytics Copilot: Ask Questions About Your Data

Ben Blake & Tina Nguyen
University of Missouri–Kansas City

Problem Statement

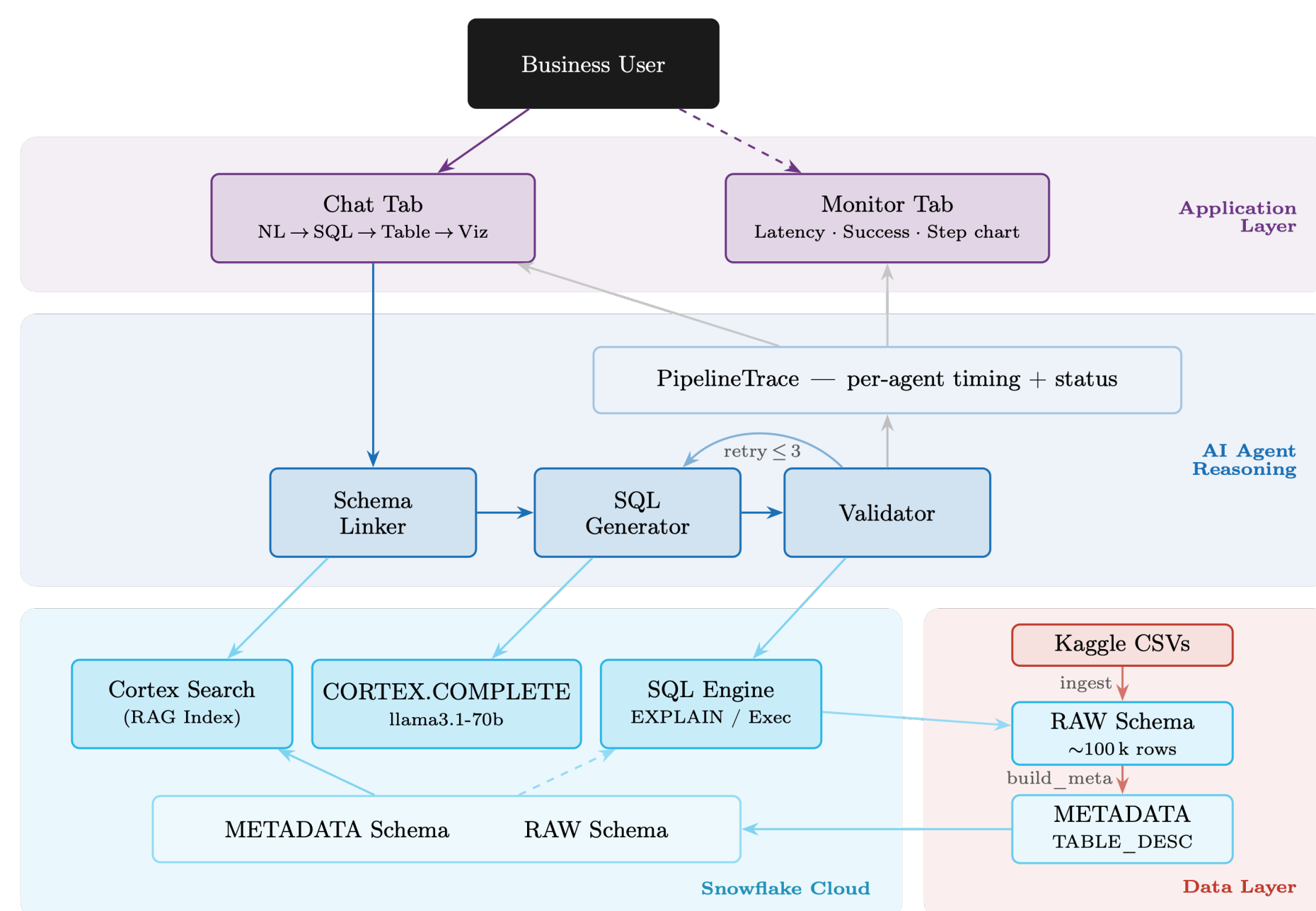


Pain Point
Non-technical users depend on analysts for SQL queries, creating bottlenecks and slowing decisions.

Solution
Three-agent AI pipeline translates natural language to SQL, executes it, and auto-visualizes results (no SQL required).

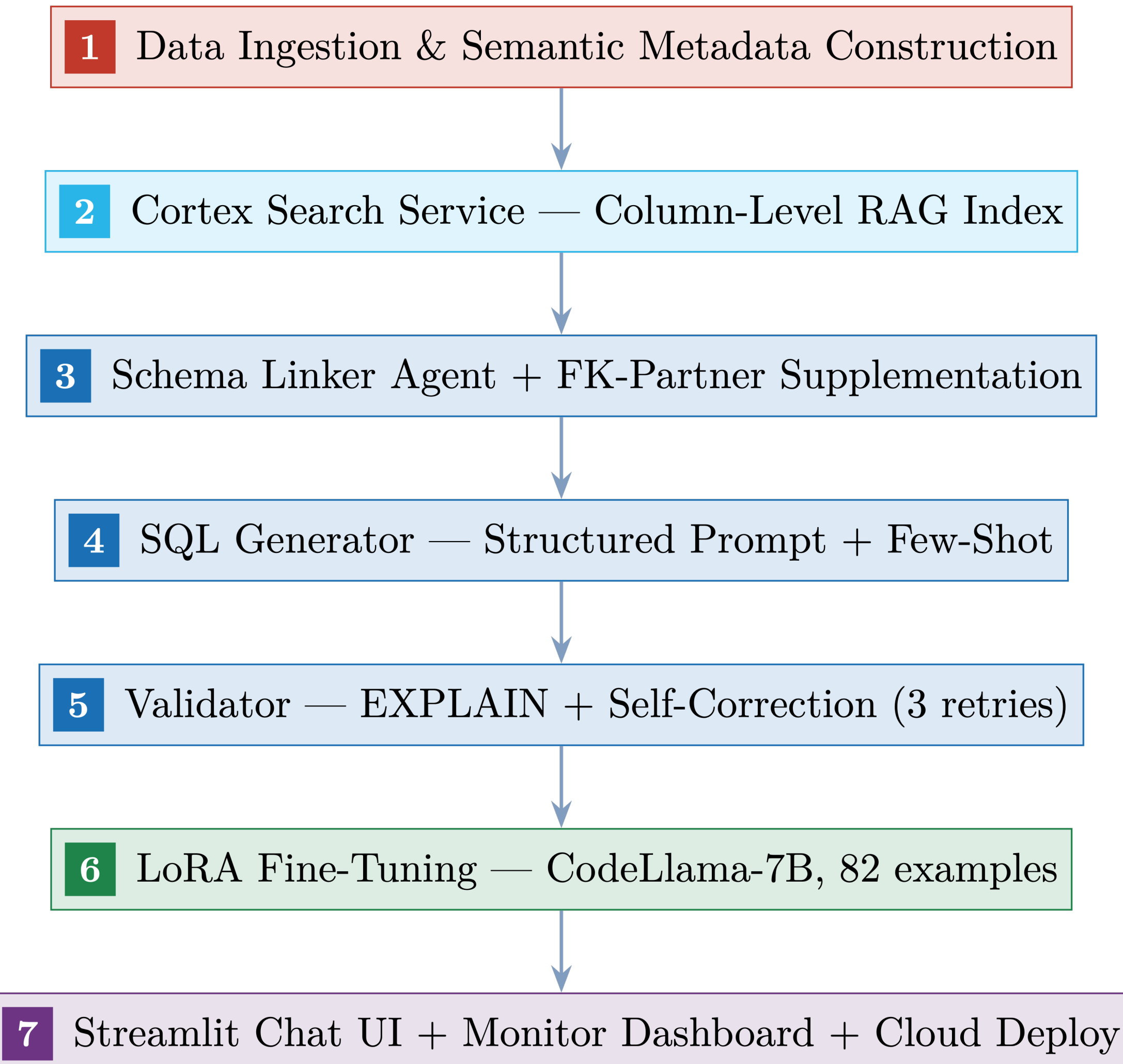
Domain
E-commerce analytics on Snowflake Cortex + Olist Brazilian dataset (100k orders, 9 tables).

System Architecture



- 1) A user query enters at the top and flows through four instrumented layers.
- 2) The **Schema Linker** narrows 9 tables to the relevant subset via semantic RAG.
- 3) The **SQL Generator** produces Snowflake SQL using structured few-shot prompting.
- 4) The **Validator** self-corrects via EXPLAIN before execution.

Methodology



Knowledge Base Construction

The nine Olist CSV tables are ingested into Snowflake’s RAW schema via a scripted Snowpark pipeline with full type coercion and PK/FK constraints. A semantic metadata layer is then built by prompting llama3.1-70b to generate natural-language descriptions, synonyms, and sample values for every column, producing a METADATA.TABLE_DESCRIPTIONS table that is indexed by a Cortex Search Service for column-level semantic retrieval.

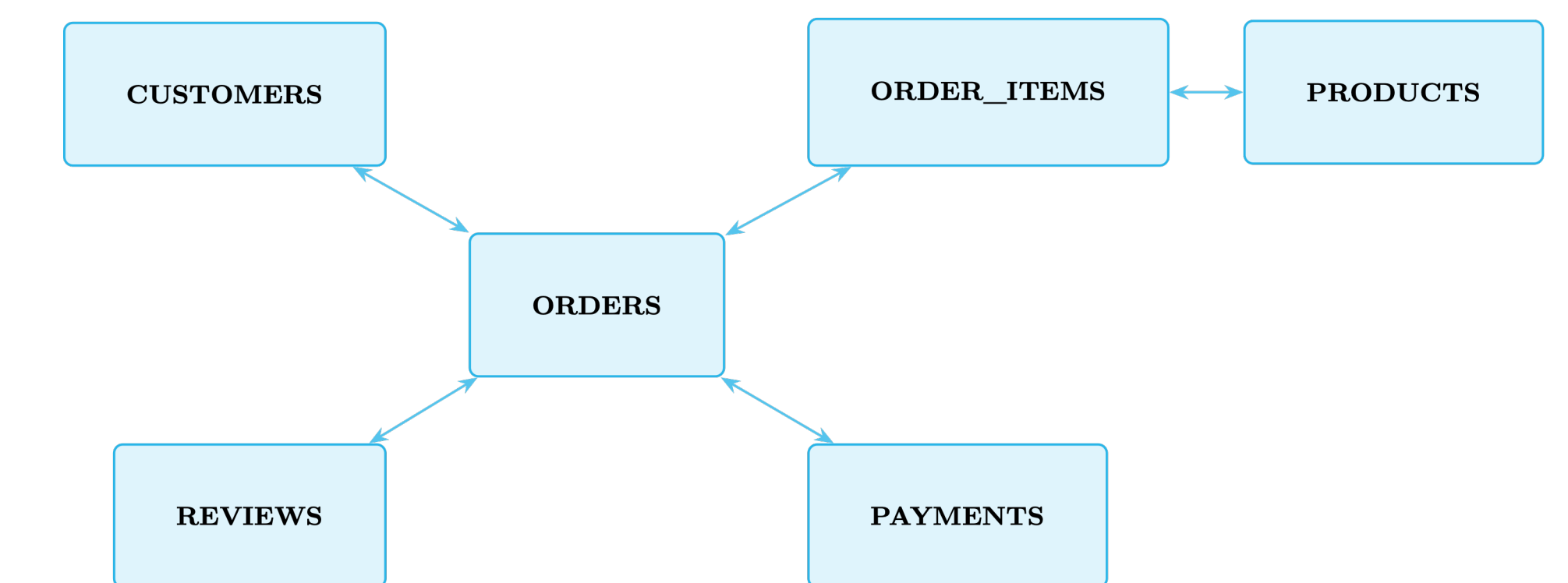
AI Reasoning Pipeline & Delivery

At query time, the Schema Linker retrieves the top-k relevant tables via semantic RAG and supplements them with FK join partners, so the SQL Generator never operates on an incomplete schema context. The Generator constructs Snowflake SQL using a 15-rule structured prompt with few-shot golden examples; the Validator executes an EXPLAIN check and feeds any errors back for up to three self-correction retries. A LoRA-adapted CodeLlama-7B (82 Olist-specific training examples, QLoRA 4-bit) provides a locally-deployable alternative inference path achieving ~10% faster generation than the zero-shot baseline. Final results are surfaced through a Streamlit chat interface with auto-visualization and a real-time monitoring dashboard, deployed to Streamlit Community Cloud.

Dataset Description

Olist Brazilian E-Commerce · Kaggle · CC BY-NC-SA 4.0

Snowflake Table	Rows	Role
ORDERS	99,441	Order lifecycle (anchor)
CUSTOMERS	99,441	Demographics, location
ORDER_ITEMS	112,650	Line items, pricing
ORDER_PAYMENTS	103,886	Payment method/value
ORDER_REVIEWS	99,224	Star ratings, comments
PRODUCTS	32,951	Category, dimensions
SELLERS	3,095	Seller location
GEOLOCATION	1,000,163	Lat/lon by zip code
CATEGORY_TRANS.	71	English category names
Total	~1.6M	2016–2018

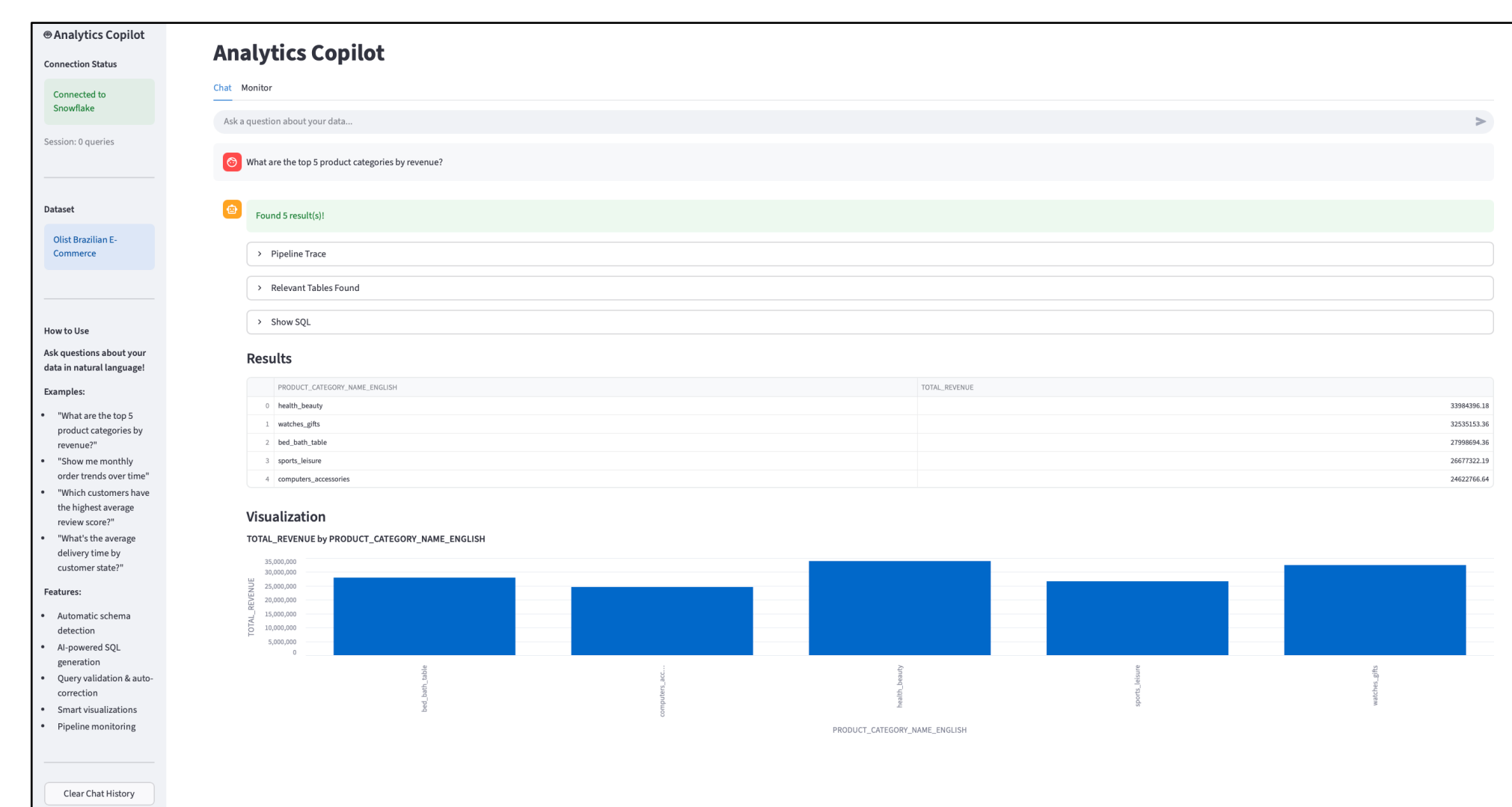


6 of 9 core tables shown

Semantic metadata layer in METADATA.TABLE_DESCRIPTIONS: LLM-generated column descriptions, synonyms, and sample values (indexed by Cortex Search for retrieval).

Application

Live Demo: `cs5542-analytics-copilot.streamlit.app`



Chat: NL query → SQL → Table → Auto-viz

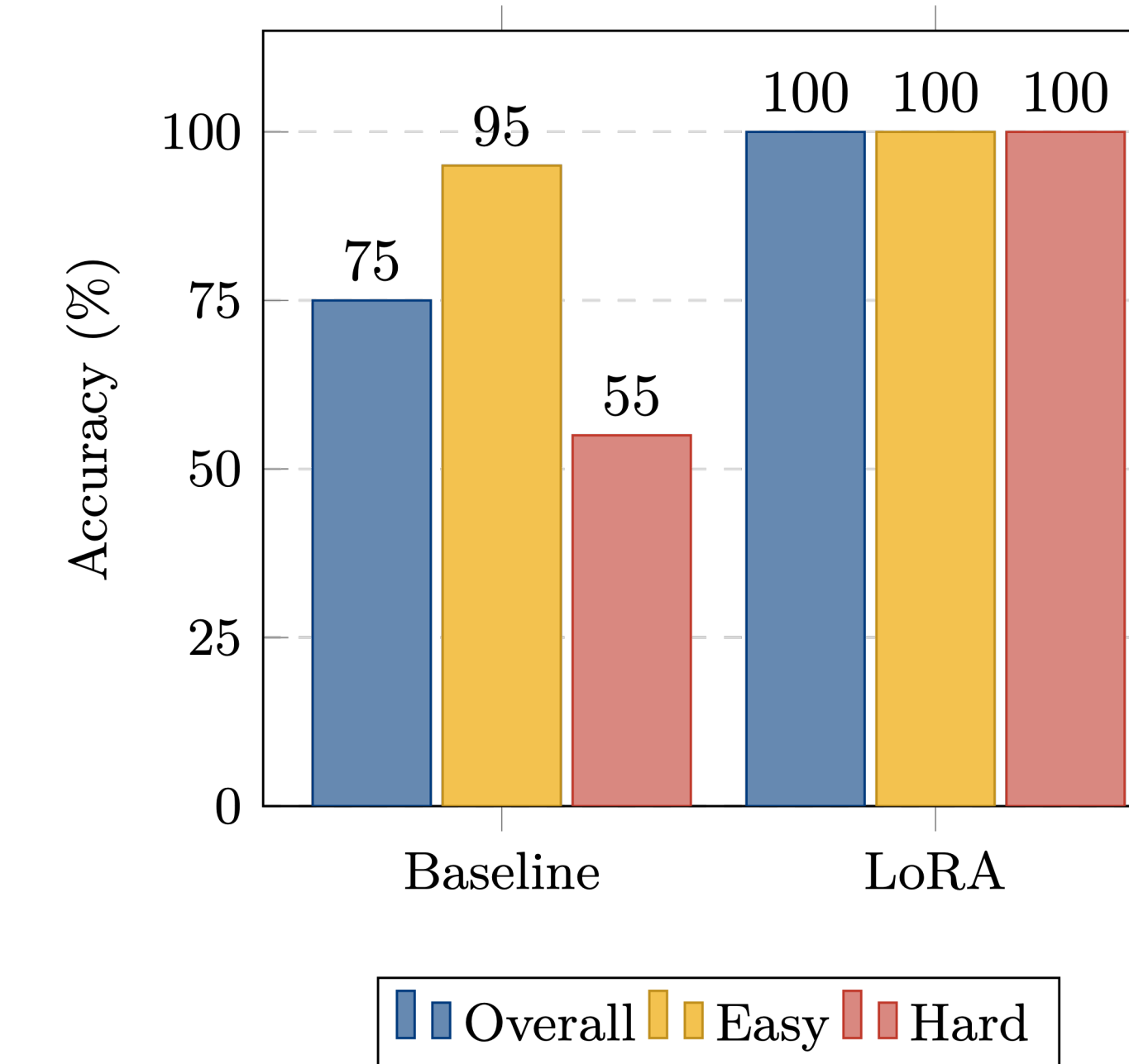
Trace: Per-agent timing, retry detection, inline

Monitor: Success rate, latency chart, step breakdown

Deploy: Streamlit Cloud; demo mode without credentials

Results & Evaluation

Untrained Baseline vs. LoRA



Phase	N	Acc.	Lat.	Notes
Baseline	50	~75%	8.0 s	Prompt eng. only
LoRA	50	100%	7.4 s	Domain-adapted

LoRA: 82 Alpaca examples

- rank $r = 16$, $\alpha = 32$
- NF4 QLoRA
- 6.5M trainable params
- ~15 min on Colab T4 GPU

Conclusion & Future Work

Delivered

- 3-agent Text-to-SQL pipeline (NL → SQL → Result)
- 100% accuracy on 50 golden queries
- LoRA domain adaptation (~10% faster inference)
- Deployed to Streamlit Community Cloud
- Single-command reproducibility (`reproduce.sh`)

Future Work

- Multi-dataset support (any Snowflake schema)
- Feedback loop to continuously improve golden queries
- Voice input + richer chart types (maps, heatmaps)
- Fine-tune on Spider/BIRD benchmark (cross-domain)
- Enterprise auth & multi-tenant deployment

Related Works

- X-SQL (NeurIPS 2025, arXiv:2509.05899)
- Chatting With Your Data (NeurIPS 2025)
- OmniSQL (NeurIPS 2025, arXiv:2503.02240)